

K8090/VM8090 for Visual Basic .NET

Developers Guide

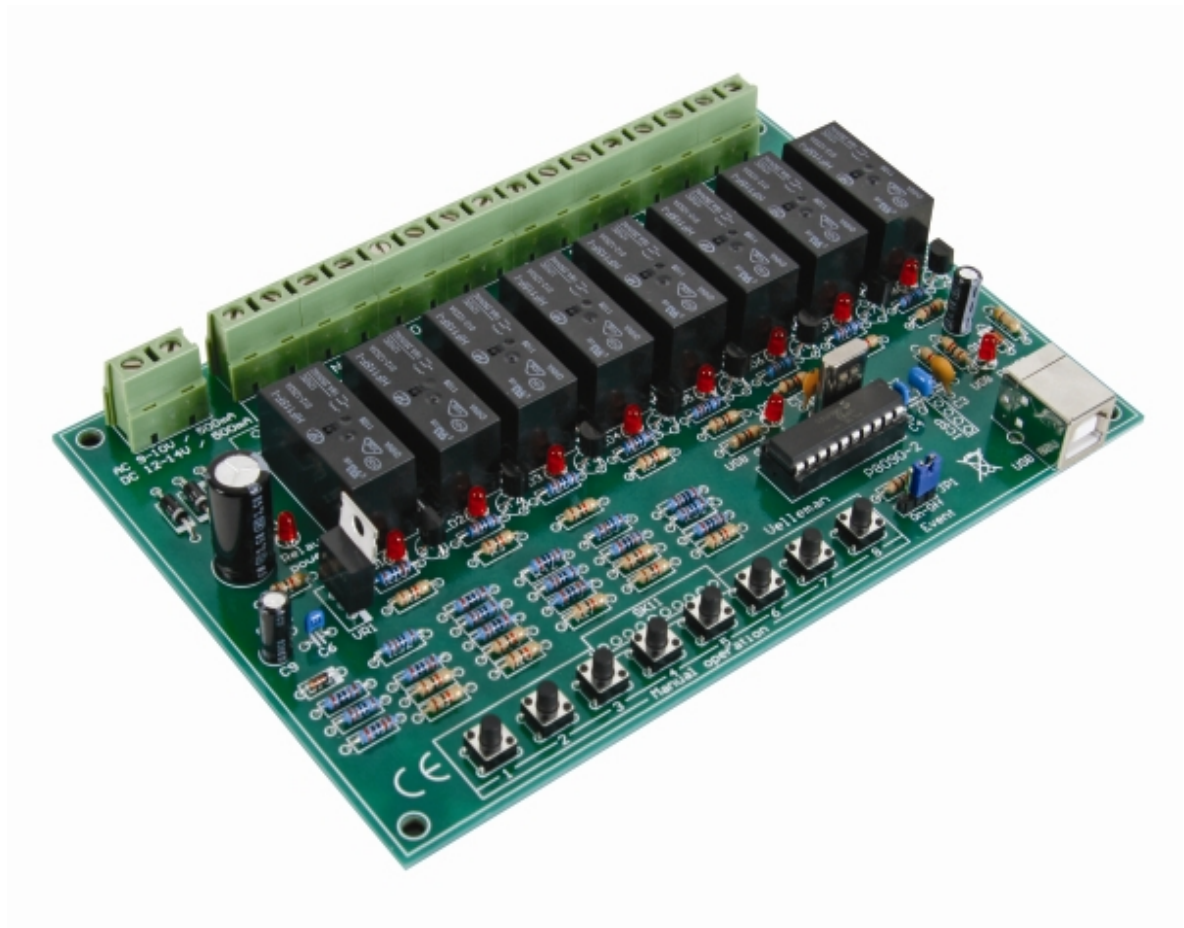


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Introduction

This guide will show you how to control the K8090 in Visual Basic .NET 2008. Readers of this guide are expected to have some minor programming experience. The VM8090 is the pre-built version of the K8090, so this guide also applies to the VM8090.

You will need to download and install Microsoft **Visual Basic 2008 Express Edition** to be able to use the examples provided. This compiler is completely free for non-commercial use.

<http://www.microsoft.com/express/Downloads/#2008-Visual-Basic>

K8090 Component for .NET

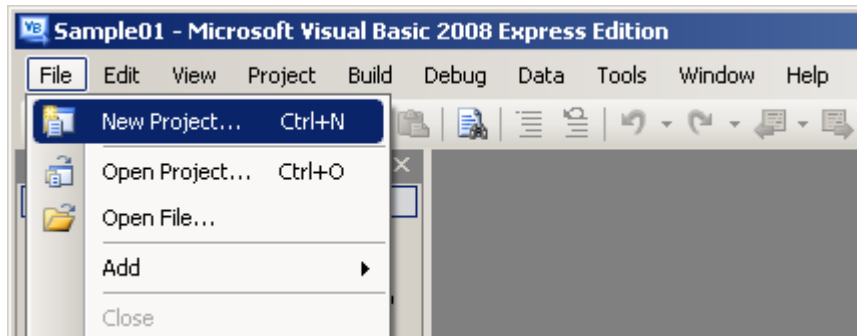
The installation comes with a K8090 component for .NET. This component simplifies the task of communicating with the K8090 in .NET (C# or VB.NET). All samples in this guide use this component. The first sample (Sample01) explains how to install (page 18) and use this component.

Samples

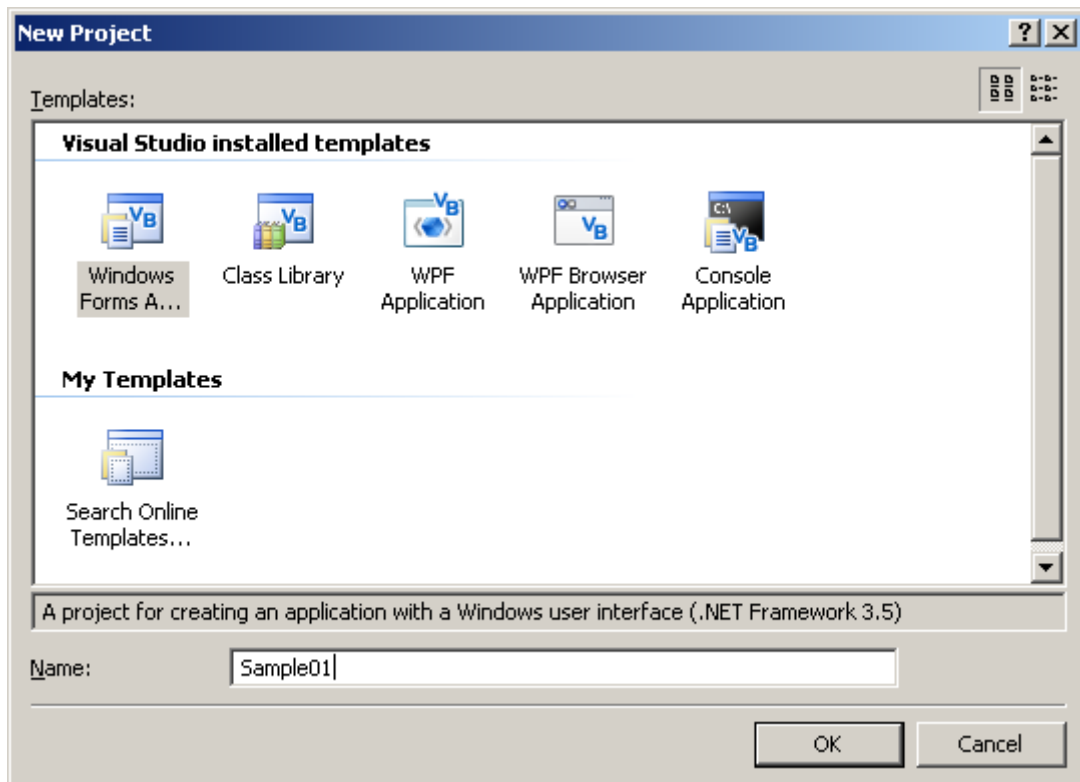
Sample 01 - Basics

This example shows the basics of connecting to the board and toggling some relays. This will give you a feel for how to use the K8090 in VB.NET and get you going for the next examples.

First, start up Microsoft Visual Basic 2008 Express Edition and create a new project by selecting **File > New Project..** from the main menu.

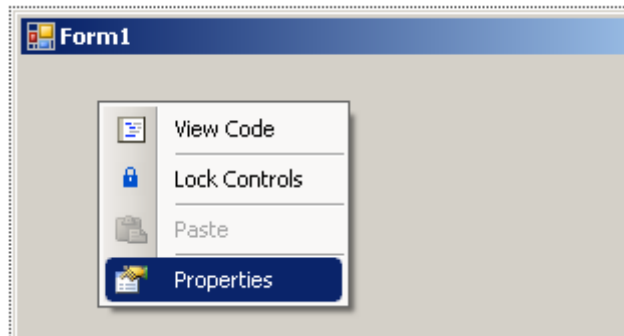


Select '**Windows Forms Application**' from the new project wizard and name it '**Sample01**' like indicated in the following picture. Click **OK** to create your project.

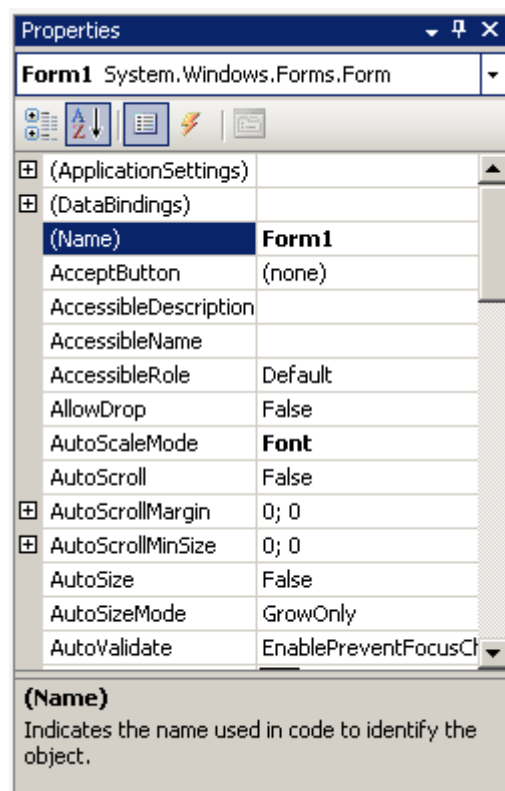


Congratulations! You have created a new Windows Forms project named 'Sample01'. You should now see an empty window (also called form) named Form1.

This is not a very suitable name for a form, so we're going to rename it to 'MainForm'. Right-click the form and select **Properties**.



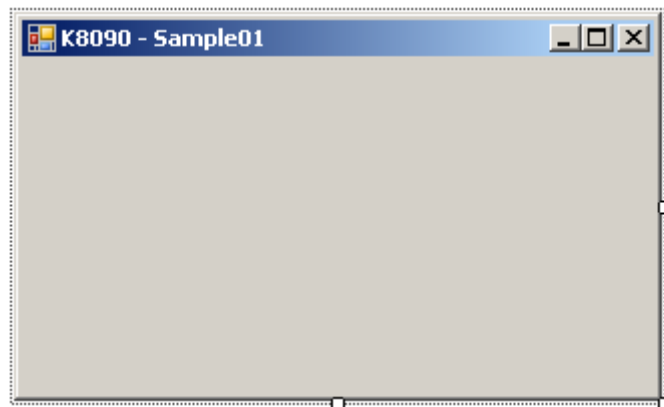
This will open up the properties window for that form.



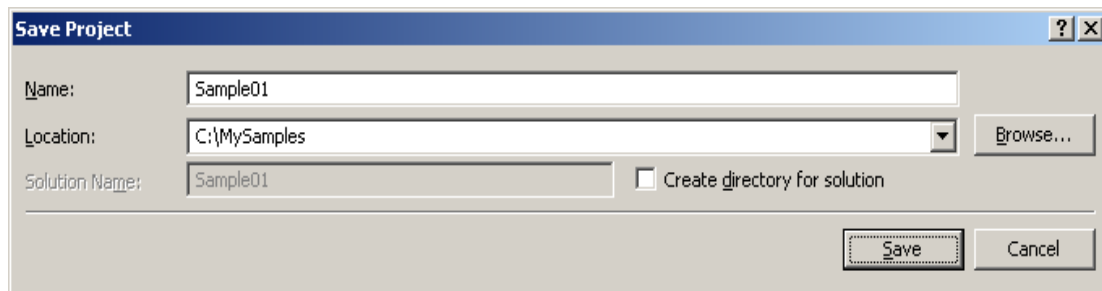
Change the following properties:

Name	MainForm
Text	K8090 - Sample01

Your form should now look something like this:



We're ready to save our project, select **File > Save All** from the main menu.

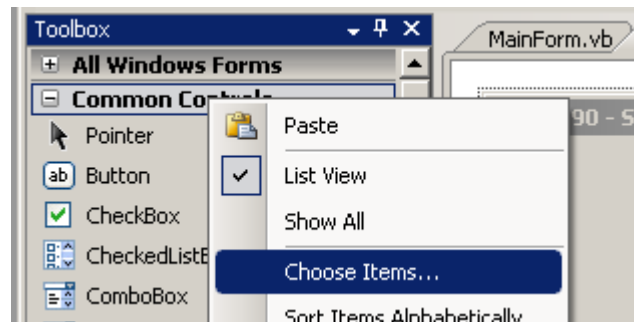


Enter the information as indicated above and press **Save**. You can select any folder under Location that suits your own preference.

Then select **File -> Save Form1.vb As..** and save your form as **MainForm.vb**.

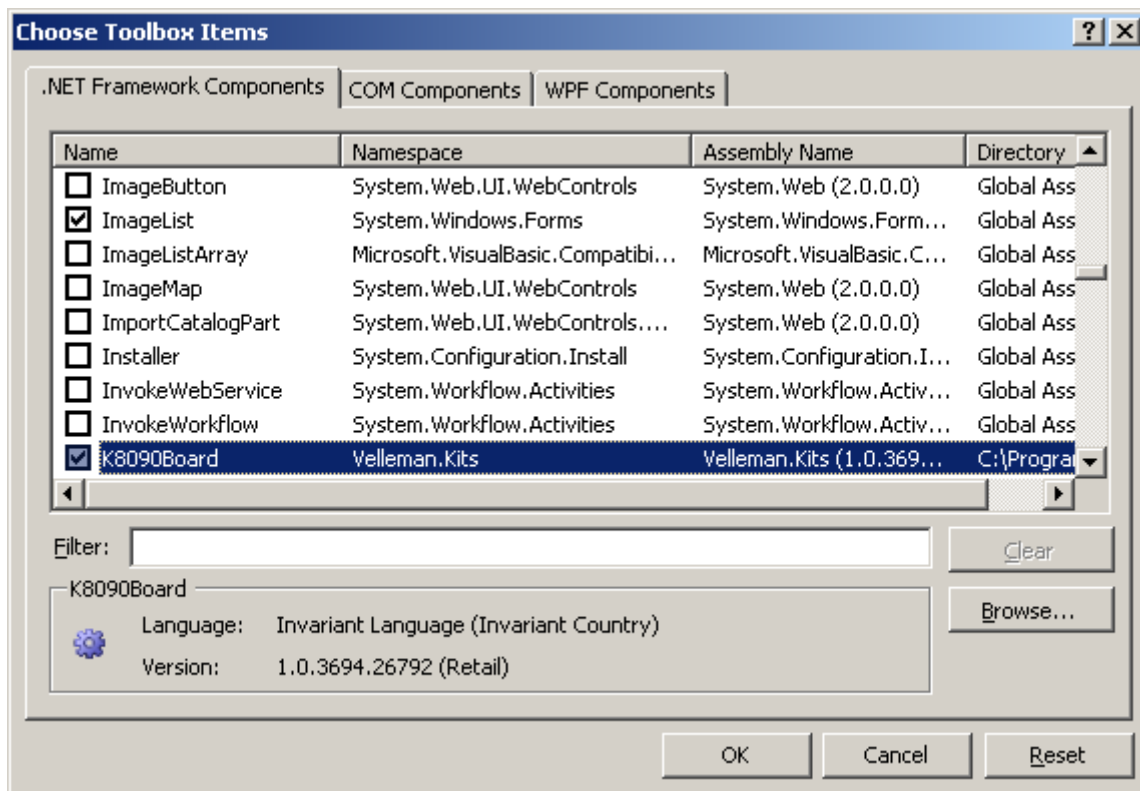
You now have a basic Windows Forms project that you can use as a template for other samples. In the next step we will add the K8090 component so we can start playing around with the board.

Open the Toolbox by clicking **View > Toolbox** from the main menu. Then right click the **Common Controls** category and click **Choose Items...**



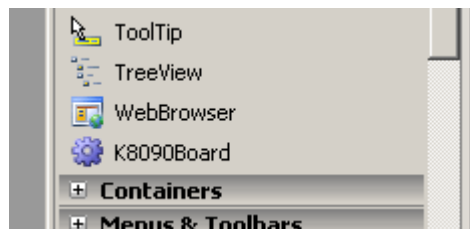
Then click Browse in the .NET Framework Components tab and navigate to the **Velleman.Kits.dll** library file in your installation directory.

By default this path is: C:\Program Files\Velleman\K8090\Kits.NET\Velleman.Kits.dll



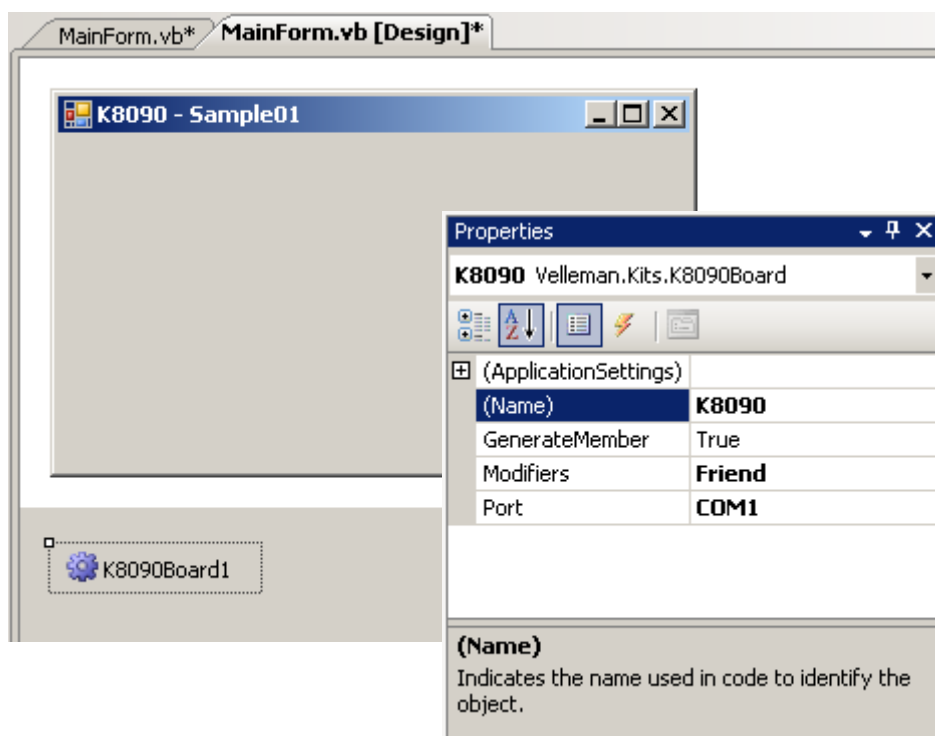
A new line should be added to the list with the name **K8090Board**. Check this item to add it to your toolbox. Click OK to continue.

You should now have a new item in your toolbox named K8090Board.



You can add this component to your project by double-clicking it or dragging it to your form. Do this now to add this component to your project.

A new item called K8090Board1 should appear at the bottom of your project.

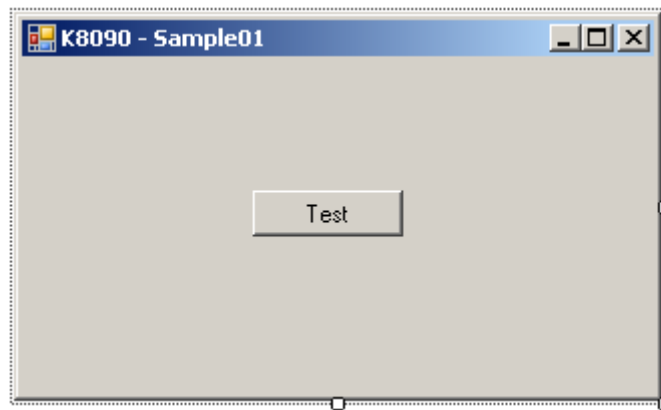


Right-click this new item, select **Properties** and rename it to 'K8090'.

Next, drag a button to your form and change the following properties:

Name	btnTest
Text	Test

Your form should now look like this:



Double-click the button to add a click event to it. The editor should open up to allow you to write code that will execute once the button has been clicked.

Type the following code:

```

MainForm.vb*   MainForm.vb [Design]*
(General)
Public Class MainForm
    Private Sub btnTest_Click(ByVal sender As System.Object)
        K8090.Port = "COM1"
        If K8090.Connect () Then
            K8090.ToggleRelay (&HFF)
            K8090.Disconnect ()
        End If
    End Sub
End Class
(Declarations)

```

This code will connect to the K8090 using port COM1, toggle all relays, and disconnect. You will need to change "COM1" to the port your K8090 is connected to!

Press **F5** to run your project. All relays on the K8090 should be toggled everytime you press the button.

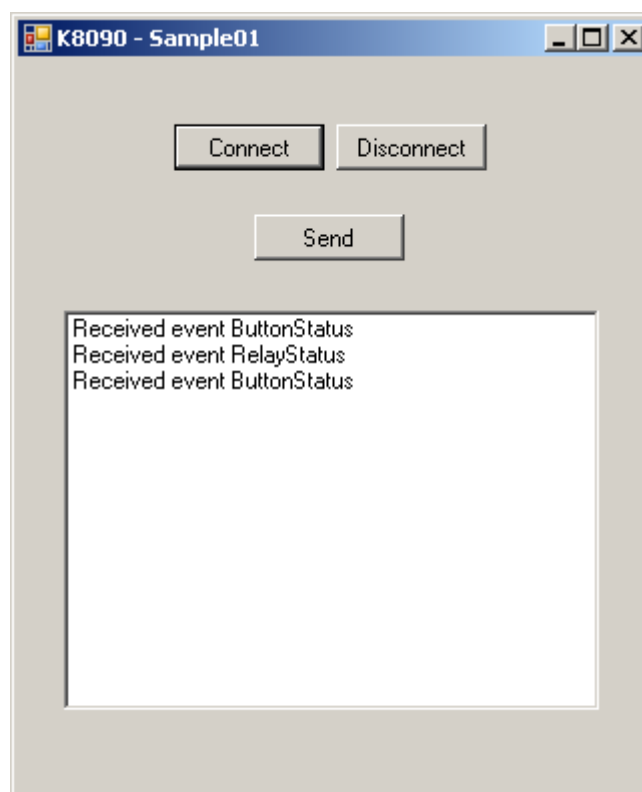
You can get the complete example by clicking Start > Programs > Velleman > K8090 > Samples > Visual Basic 2008 Express Edition > **Sample01**.

Sample 02 - Event handling

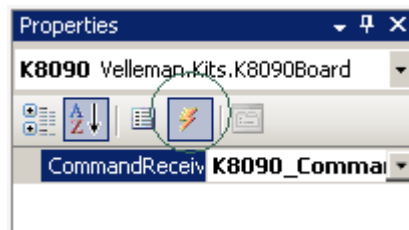
In this chapter we will be extending the first example with some event handling. We will try to capture the events sent by the board.

To start, open the Sample02 project located in your installation's samples folder. Or select Start > Programs > Velleman > K8090 > Samples > Visual Basic 2008 Express Edition > **Sample02**.

You should see a form with three buttons and a listbox. Press Connect to connect to your K8090 and then press a button on your K8090, you should see three or more events that indicate that you **pressed** (and **released**) a button. If the state of one or more relays changed when you pressed this button, you should also see a **relay status** event.



So how does this all work? Well, every time the board sends a packet, it is captured by our K8090 component which fires an event. You can see this event by opening the event tab in the properties window.



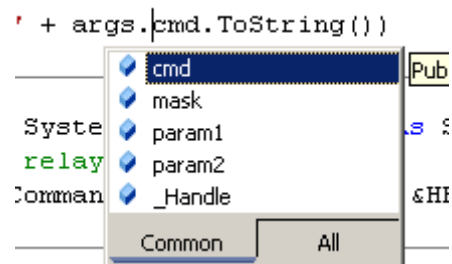
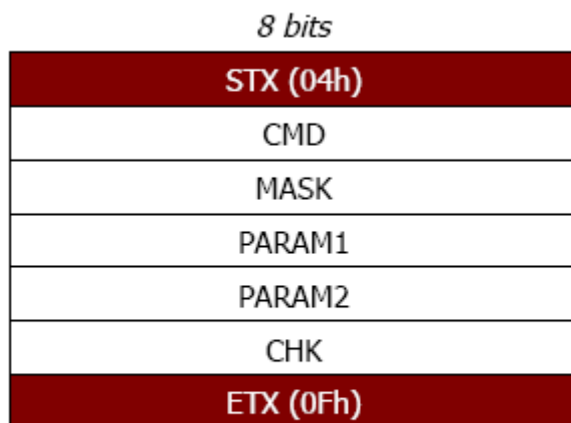
This event simply adds the name of the event to the listbox everytime an event is fired. Let's take a look at the code. Our **CommandReceived** event handler is where the magic happens:

```
Private Sub K8090_CommandReceived(ByVal o As System.Object, _
    ByVal args As Velleman.Kits.CommandEventArgs) _
    Handles K8090.CommandReceived

    ' Add the event to our list
    PacketList.Items.Add("Received event " + args.cmd.ToString())
End Sub
```

The **args** parameter contains all information on our event (cmd, mask, param1, param2). This information is identical to the information described in the **Protocol Manual**.

Open this manual by clicking Start > Programs > Velleman > K8090 > Help > K8090 Protocol Manual (PDF). Browse to the chapter '**Packet Composition**' on page 2



1. K8090 Packet Diagram

As you can see, the args parameter contains all information you need as described in the protocol. You should get familiar with this protocol.

Events can also be initiated by the user. You could for example want to switch all the relays ON. If the relays were switched off, then an event will be sent notifying you that all relays have now been switched on. Let's give this a try. Open the **Click** event of the **Send** button (btnSend).

```
Private Sub btnSend_Click(ByVal sender As System.Object, _
    ByVal e As System.EventArgs) Handles btnSend.Click

    K8090.SendCommand(
        Velleman.Kits.K8090Command.SwitchRelayOn,
        &HFF,
        &H0,
        &H0
    )

End Sub
```

We are using the **SendCommand** function sends a raw command to the K8090 like described in the Protocol Manual on page 4. Note that the command we send here is identical to the command we sent in our first example:

```
K8090.SwitchRelayOn(&HFF)
```

The SwitchRelay method also issues a SendCommand behind the screens. The mask parameter has been set to &HFF to indicate we want to switch all relays on.

In Visual Basic .NET the '&H' prefix tells the compiler that we are using hexadecimal notation. So &HFF is the hexadecimal value FF. Each bit of the mask parameter corresponds to a relay: bit 1 is relay 1, bit 2 is relay 2, and so on.

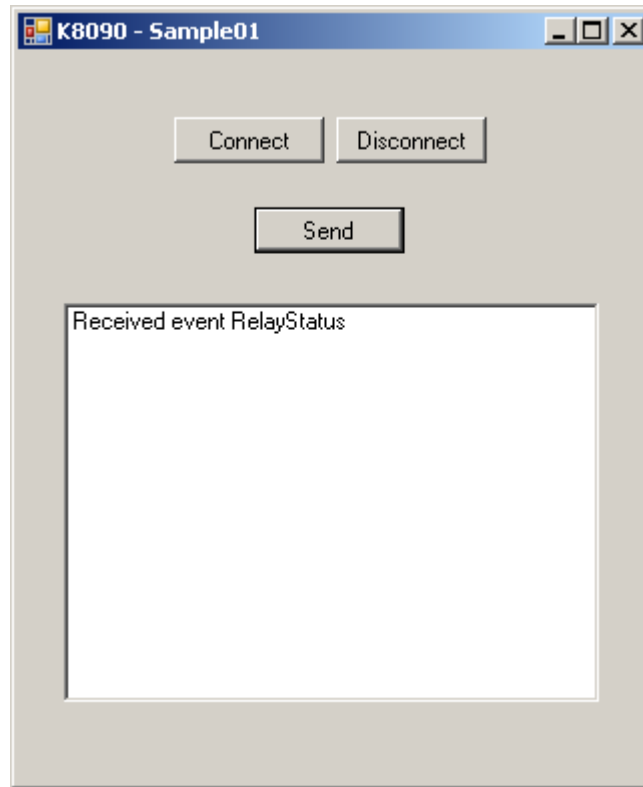
	1 Byte							
Bits	8	7	6	5	4	3	2	1
Relay	Relay 8	Relay 7	Relay 6	Relay 5	Relay 4	Relay 3	Relay 2	Relay 1

So, suppose we want to activate relays 3 and 8, we would get the following:

	1 Byte							
Bits	8	7	6	5	4	3	2	1
Relay	Relay 8	Relay 7	Relay 6	Relay 5	Relay 4	Relay 3	Relay 2	Relay 1
Binary	1	0	0	0	0	1	0	0

So we get the binary value 10000100 = &H84 which we can use as the mask parameter to switch on relays 3 and 8.

Once you send this command you should also get a 'Relay status (51h)' event as described on page 11 in the Protocol Manual. The event handler of our application should list it in our ListBox as a new line with the name 'Received event RelayStatus'.



Experiment!

You should take your time to get to know the board and what it can do. Experiment with all the different commands and most importantly, have fun!

I hope you enjoy the K8090, check for updates regularly on our website.